

GEO systematic search (Sep 2026): human ascending/thoracic aortic aneurysm and dissection tissue transcriptomes

ATAA vs non-diseased aorta
4 bulk cohorts, n = 115
GSE26155 (53) . GSE140947 (12)
GSE235161 (42) . GSE202267 (8)

TAAD vs non-diseased aorta
6 bulk cohorts, n = 76
GSE52093 (12) . GSE153434 (20) . GSE98770 (11)
GSE294606 (13) . GSE267434 (12) . GSE147026 (8)

Independent validation resources
Direct contrast: GSE318877 (TAAD vs ATAA media, 70 punches)
scRNA-seq: GSE155468 (ATAA) + GSE213740 (TAAD), 73,755 cells
Excluded after QC: GSE190635 (label inconsistency)

Harmonised per-cohort processing (symbol harmonisation, re-processing from raw where needed) -> per-cohort DE (limma-trend for arrays/FPKM; DESeq2 for counts) -> QC: PCA, cross-cohort concordance of DE profiles

Random-effects meta-analysis (Stouffer + DL)
ATAA consensus DEGs; LOCO robustness

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Validation

- Direct TAAD-vs-ATAA contrast (GSE318877): gene-level, pathway-level, signature scores
- scRNA-seq: cell-type-resolved pseudo-bulk DE, cellular source of gene classes, composition
- LOCO classifier AUC across 10 cohorts

ATAA vs TAAD comparison
Gene classes (shared / TAAD-specific / ATAA-specific / discordant; z_diff) . GSEA on meta-rankings (Hallmark, KEGG, Reactome) and ORA
ssGSEA cell-composition inference (aorta scRNA-derived markers) . STRING PPI hubs . control-referenced signature scores + LOCO classifier

Output: distinct and shared molecular programmes of ascending aortic aneurysm and acute type A dissection